Grade 11 Global Economics — Term 1 Review Outline

2026–2027

1 Introduction

This term starts with choosing a probability model and finding probabilities and theoretical statistics. You then use sample records to estimate probabilities, summarize data, and compare the results with the model. Normal and Student's t tables let you move between cutoffs and probabilities. In C10, you bring modelling and data analysis together to investigate an economic or financial question.

Use the summary to match a question to a method and recall what to write. Follow a criterion link when you need to see the steps or check your working against an example.

A routine for any question. (1) Define the variable and list the givens, with units. (2) Mark the event or statistic requested and choose a model or data method. (3) Write the formula, counts, area, or table entry and substitute the values. (4) Check the support, units, and rounding, then explain what the result means in the situation.

Rounding. For C1–C7, report probabilities to four decimals, percentages to two, and statistics, rates, and times to three where needed. Keep unrounded values for comparisons. For C8–C9, follow the rounding steps beside the table methods.

2 Summarized Outline

	Question cue	**Choose**	**Write on paper**
C1	Count, measurement, or model description	Discrete or continuous family	Variable, clue, parameters, support, and meaning of any area
C2	Flat or straight-sided density	Rectangle, triangle, or trapezoid areas	Support, height, lines if needed, shaded bounds and area
C3	Repeating cycle or uniform heading	Length or angle proportion	Pieces across the restart, favorable measure, full measure
C4	Wait until next arrival or infer rate	Exponential tails and Poisson count mean	Rate units, event equation, logarithm if solving for rate
C5	Model mean, variance, or SD	Theoretical moment formulas	Parameters, substitutions, units, interpretation
C6	Observed fraction in an event	Relative frequency and model errors	Endpoint rule, $k, n, k/n$, and errors
C7	Mean or spread from records	Weighted sample statistics	Frequencies, values or midpoints, sums and $n-1$
C8	Probability or cutoff with a t model	Printed one-tail t table	df, signed cutoffs, columns, tail arithmetic or back-transform
C9	Compare normal and t	Both lookups at the shared scale	Both results, absolute difference and two percentage differences
C10	Propose a model or investigate a question	Match a model to real economic data	Brief choice for the exam, methods and evidence for the report

Model clues and supports

Clue	Model and parameters	Support
One success/failure outcome	Bernoulli(p)	$\{0, 1\}$
Successes in n independent trials with common p	Bin(n, p)	$\{0, 1, \ldots, n\}$
Independent trials through first success with common p	Geom(p)	$\{1, 2, \ldots\}$
Count in a period of length d in a Poisson process	Poisson(λd)	$\{0, 1, 2, \ldots\}$
Equal-length intervals equally likely on a bounded range	$U(a, b)$	$[a, b]$
Straight slopes with a peak at c, possibly at an endpoint	Tri(a, b, c)	$[a, b]$
Exact normal bell centered at μ with SD σ	$N(\mu, \sigma^2)$	$\mathbb{R}$
Waiting time to the next event in a Poisson process	Exp(λ)	$[0, \infty)$

Theoretical moments

Model	Mean μ	Variance σ^2	SD
$U(a, b)$	$(a + b)/2$	$(b - a)^2/12$	$(b - a)/\sqrt{12}$
Tri(a, b, c)	$(a + b + c)/3$	$(a^2 + b^2 + c^2 - ab - ac - bc)/18$	$\sqrt{\sigma^2}$
Tri(a, b, a), decreasing	$(2a + b)/3$	$(b - a)^2/18$	$(b - a)/\sqrt{18}$
Tri(a, b, b), increasing	$(a + 2b)/3$	$(b - a)^2/18$	$(b - a)/\sqrt{18}$
$N(\mu, \sigma^2)$	μ	σ^2	σ
Exp(λ)	$1/\lambda$	$1/\lambda^2$	$1/\lambda$

Reading a sample record

Record	Probability (C6)	Statistics (C7)
Raw list	Count qualifying entries, including repeats.	Use actual values with weight 1 or tally repeats.
Exact frequency table	Add frequencies of qualifying values.	Weight each exact value by its frequency.
Grouped classes	Add counts for complete eligible classes.	Use each class midpoint with its frequency.
Frequency histogram	Read bar heights as counts, then use class bounds.	Read heights as weights and use class midpoints.

Check the class boundaries. When an event cuts through a class, the grouped record alone leaves the within-class count unknown.

Choosing and reading a table

Table	Given a cutoff (direct)	Given an area (inverse)
Normal CDF	Round z to two decimals. Row and hundredths column for $\lvert z\rvert$ give $\Phi(\lvert z\rvert)$. Reflect for negative z.	Convert to a left area. Find the closest printed CDF (using symmetry for negative cutoffs), then read z.
Student's t	Set $\nu = n - 1$. Find $\lvert c\rvert$ in that row to read its one-tail column α. Reflect or complement as needed.	Choose row ν and the one-tail column α. Assign a sign and return to context with $x = m + ac$.

Use the 2026–2027 *Standard Normal Distribution and Student's t Reference Sheets.*

3 Step-by-Step Outline

C1 — Identify the variable, model, and support

Method. Use these steps when a question describes how a variable is measured or generated.

1. Define X in context. A count or coded outcome is discrete. A measurement that can take any value in an interval is continuous.
2. Match the wording to a model clue. Name the parameters and give their values with units where applicable. **Check.** A stated rate must be multiplied by the observation length to get a Poisson count mean.
3. Write the support, the possible values under the model. If the description only suggests a shape, state the assumption needed to use it.

Highlight key. Model/support · Event/area · Table lookup · Sample data/statistics · Population/model quantities

Worked example. Identify the models from these short descriptions.

Variable and clue	Classification, model, and support
One cooperative member's repayment, timely with probability 0.92	Discrete. Code timely as 1 and late as 0. Bernoulli(0.92), $\{0,1\}$.
Timely repayments among 40 independent members, each with $p = 0.92$	Discrete. Fixed number of trials gives $\text{Bin}(40, 0.92)$, $\{0,\ldots,40\}$.
Invoices checked through the first error, with independent checks and $p = 0.08$	Discrete. $\text{Geom}(0.08)$, $\{1,2,\ldots\}$. The first error counts as a trial.
Enquiries in 3 days, with independent arrivals at a constant rate of 5 per day, one at a time	Discrete. Count mean $5(3) = 15$ gives $\text{Poisson}(15)$, $\{0,1,\ldots\}$.
Scale error equally likely over equal-length intervals from -3 to 3 g	Continuous. $U(-3,3)$, $[-3,3]$ g.
Poster completion time on $[10,30]$ min with straight branches peaking at 18 min	Continuous. $\text{Tri}(10,30,18)$, $[10,30]$ min. A single rising line for the tank fraction on $[0,1]$ gives $\text{Tri}(0,1,1)$.
Filling error with an exact normal bell, center 0 g and SD 4 g	Continuous. $4^2 = 16$ gives $N(0,16)$, $\mathbb{R}$ in grams.
Time to the next repair request in a Poisson process of rate 12 per hour	Continuous. $\lambda = 12/60 = 1/5$ per minute gives $\text{Exp}(1/5)$, $[0,\infty)$ min.

Check. A steadily decreasing delivery-time pattern on $[15,45]$ min gives $\text{Tri}(15,45,15)$ if you assume a straight line. Symmetric masses centered at 5 kg with SD 1.2 kg give $N(5,1.44)$ if you assume a normal shape. In the second case the model support is $\mathbb{R}$, while real masses are nonnegative.

Method. Read a continuous density in three ways.

1. Read $f(x)$ as the height at x, in inverse units. For the scale error $Y \sim U(-3,3)$, the height is $1/6$ g^{-1} on the support and $P(Y = 0) = 0$.
2. Read $P(l < Y < u)$ as the area between the two cutoffs. For $-1 < Y < 1$, the area is $2(1/6) \approx 0.3333$.
3. Read $F(u) = P(Y \le u)$ as all the area to the left. Here $F(1) = (1 - (-3))/6 \approx 0.6667$, so about 66.67% of errors are at most 1 g. **Check.** Area is unitless. Including an endpoint leaves a continuous probability unchanged.

C2 — Find probabilities from density areas

Method. Use geometry for constant or piecewise-linear densities.

1. Mark the support and the event, then keep only the part of the event inside the support.
2. Find the height or line equations if needed. The total area is 1.
3. Split at each change of slope. Add rectangles, triangles, or trapezoids, or subtract simpler outside pieces from 1. A straight segment from l to u contributes $(u-l)[f(l)+f(u)]/2$.

Worked example. Uniform labelling time $B \sim U(3, 15)$ min. The flat height is $1/12$ min^{-1}. For 6 to 9 min, the rectangle gives $P(\,6 \le B \le 9\,) = (9-6)/12 =$ 0.2500, so one quarter of times fall in that interval.

Worked example. Market-stall setup time S in minutes. Find $P(\,9 \le S \le 19\,)$.

1. Use Tri$(5, 25, 13)$. The support is $[5, 25]$ and the mode is 13, so the vertices are $(5, 0), (13, H), (25, 0)$. Normalize with $1 = \frac{1}{2}(25-5)H$, giving $H = 1/10$ min^{-1}.
2. Find each slope using rise over run, and anchor each line at its zero endpoint.

$$m_1 = \frac{1/10}{13-5} = \frac{1}{80}, \qquad m_2 = \frac{-1/10}{25-13} = -\frac{1}{120}, \qquad f(s) = \begin{cases} (s-5)/80 & 5 \le s \le 13, \\ (25-s)/120 & 13 < s \le 25, \\ 0 & \text{otherwise.} \end{cases}$$

   Check. Both branches reach $1/10$ at the mode.

3. The event crosses the mode. With $f(9) = f(19) = 1/20$, remove the two outside triangles:

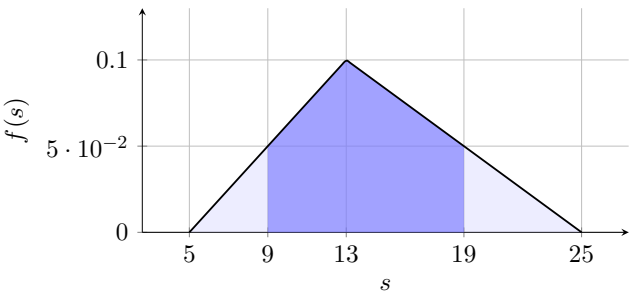
$$1 - \tfrac{1}{2}(9-5)(1/20) - \tfrac{1}{2}(25-19)(1/20) = 0.7500\,.$$

   Equivalently, the two inside trapezoids give $\frac{1}{2}(4+6)(1/20+1/10) = 0.7500$. The model puts 75.00% of setup times between 9 and 19 min.

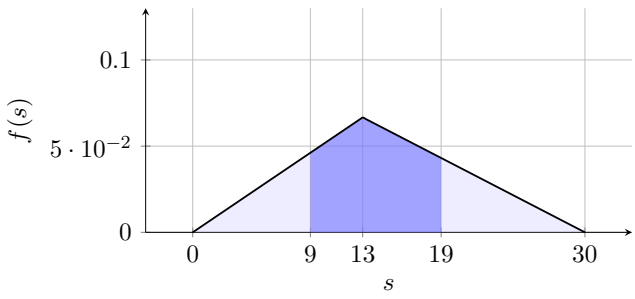
Changed support. With support $[0, 30]$, keep mode 13. Recalculate the height $2/30 = 1/15$, the slopes $(1/15)/13 = 1/195$ and $-(1/15)/17 = -1/255$, and the branch heights at the cutoffs. The new lines are $s/195$ on $[0, 13]$ and $(30-s)/255$ on $(13, 30]$, with zero density elsewhere. The same event now covers

$$1 - \tfrac{1}{2}(9)(9/195) - \tfrac{1}{2}(11)(11/255) = \frac{368}{663} \approx 0.5551,$$

or 55.51% of setup times. **Check.** The bases and heights of both removed triangles changed.



Original. Read the darker area from 9 to 19 min under the peak at 13 min.



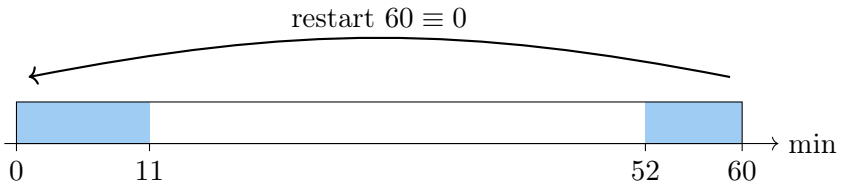
Changed support. Read the same cutoffs under the lower peak. Both vertical axes show density in min^{-1}.

C3 — Handle wraparound in cycles and directions

Method. Use length for a uniform position in a cycle and angle for a uniform direction.

1. Mark the full cycle or turn. Put the cutoffs and the full measure in the same units.
2. If the event crosses the restart or north, split it into the piece before and the piece after that point. **Check.** A tolerance *on either side* contributes twice.
3. Add the favorable lengths or angles, then divide by the length of the full cycle or angle of the full turn.

Worked example. An alarm is uniform in a 60-minute inspection cycle. From minute 52 through the restart to minute 11, the event is $[52, 60) \cup [0, 11]$.

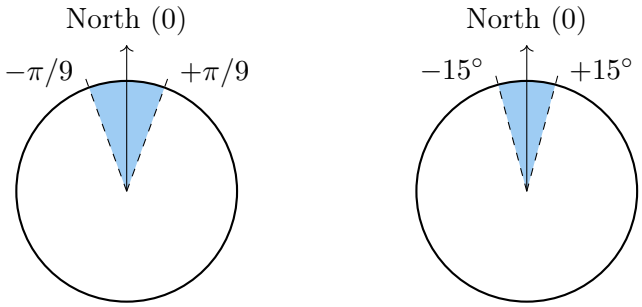


Add the two blue lengths. The arrow connects minute 60 to minute 0.

$$P = \frac{(60 - 52) + 11}{60} = \frac{19}{60} \approx 0.3167.$$

About 31.67% of alarms fall in that window. For the variant within 200 seconds of a restart, $200/60 = 10/3$ min per side. The event is $[170/3, 60) \cup [0, 10/3]$, so $P = 2(10/3)/60 = 1/9 \approx 0.1111$. About 11.11% of alarms fall within that tolerance.

Worked example. A drone's heading is uniform around north. Compare two angular tolerances.



Read the whole blue sector, on both sides of north. A full turn is 2π radians or $360°$.

$$[0, \pi/9] \cup [17\pi/9, 2\pi) \text{ rad} \implies P_{\pi/9} = \frac{2(\pi/9)}{2\pi} = \frac{1}{9} \approx 0.1111,$$

$$[0, 15] \cup [345, 360) \text{ degrees} \implies P_{15°} = \frac{2(15°)}{360°} = \frac{1}{12} \approx 0.0833.$$

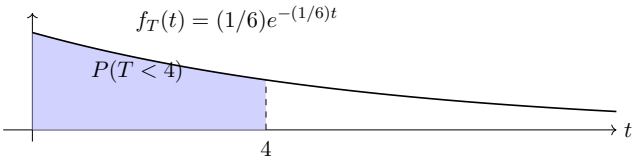
The $\pi/9$ tolerance is more likely by $1/9 - 1/12 = 1/36$, or 2.78 percentage points. Relative to the $15°$ probability, that is $(1/36)/(1/12) \times 100\% = 33.33\%$ higher.

C4 — Model waiting times and recover rates

Method. Use Exp(λ) for the wait T to the next event in a Poisson process.

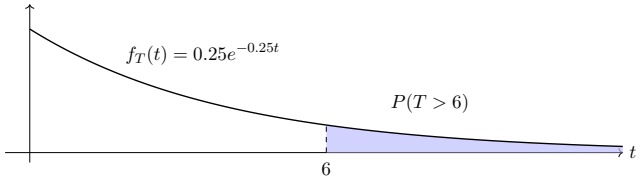
1. Check that arrivals are independent, occur one at a time at a constant rate over the relevant period, and that multiple arrivals in a very short interval are negligible. Use a separate rate for each stable shift if rates change. A wait that spans a rate change needs a different model.
2. Use rate = expected count divided by elapsed time, or $\lambda = 1/E(T)$. **Check.** The rate and cutoff units must cancel in λt. The density is $f_T(t) = \lambda e^{-\lambda t}$ on $[0, \infty)$, zero elsewhere.
3. Mark the requested region. For $t \geq 0$, the area right of t is $R(t) = e^{-\lambda t}$, and the area left of t is $1 - R(t)$. Subtract nested areas for an interval and add disjoint areas for outside tails.

Worked example. Charging spaces become free at a constant rate within each scenario. Read t in minutes and $f_T(t)$ in inverse minutes.



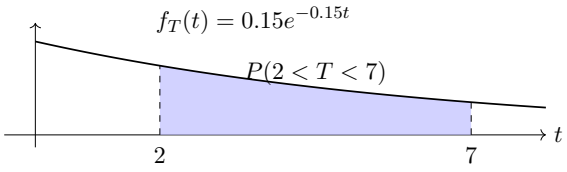
Left of 4. Rate $5/30 = 1/6$ per min.

$P(T < 4) = 1 - e^{-4/6} \approx 0.4866$.



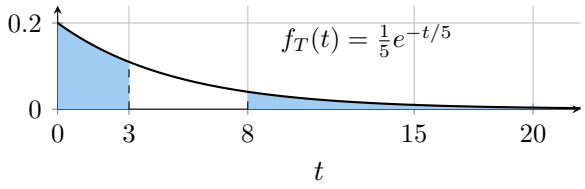
Right of 6, continuing to infinity. Mean wait 4 min gives rate 1/4 per min.

$P(T > 6) = e^{-6/4} \approx 0.2231$.



Between 2 and 7. Rate 3/20 per min.

$P(2 < T < 7) = e^{-3/10} - e^{-21/20} \approx 0.3909$.



Before 3 or after 8. Rate $12/60 = 1/5$ per min. The right tail continues to infinity.

$P(T < 3 \text{ or } T > 8) = 1 - e^{-3/5} + e^{-8/5} \approx 0.6531$.

The four modeled chances are 48.66%, 22.31%, 39.09%, and 65.31% for their respective opening windows.

Method. Work backward from a waiting probability to find an arrival rate.

1. Convert the given probability to a right-tail area r. Take logs in $e^{-\lambda t} = r$, giving $\lambda = -\ln(r)/t$.
2. Express rates in the same time unit. Multiply a per-minute rate by 60 for a per-hour rate, keeping the exact value during conversion.

Worked example. Recover rates from tails. In a morning receipt shift, $P(T > 10) = 0.40$. Then $\lambda = -\ln(0.40)/10 \approx 0.092$ receipts/min, giving $60\lambda = -6\ln(0.40) \approx 5.498$ receipts/hour. In the afternoon $P(T < 6) = 0.80$, so $r =$ 0.20 and $\lambda = -\ln(0.20)/6 \approx 0.268$ receipts/min, giving 16.094 receipts/hour. The modeled receipt rate is higher in the afternoon.

Method. **C4 continued.** Use Poisson count means when the question compares expected counts.

1. Find both rates in the same units and write $E[N(t)] = \lambda t$ for each process.
2. Equate the difference to the requested count and solve for the common elapsed time.

Worked example. Compare expected counts for two payment channels. If $P(T_A < 5) = 0.75$, then $e^{-5\lambda_A} = 0.25$ gives $\lambda_A = \ln 4/5$. If $P(T_B > 5) = 0.50$, then $e^{-5\lambda_B} = 0.50$ gives $\lambda_B = \ln 2/5$. Both rates are payments/min.

$$E[N_A(t)] - E[N_B(t)] = (\lambda_A - \lambda_B)t = 3 \quad \Longrightarrow \quad t = \frac{15}{\ln 2} \approx 21.640 \text{ min.}$$

After that elapsed time A's expected count is three higher. **Check.** The actual counts are random.

C5 — Compute and interpret theoretical moments

Method. Use model parameters to find the expected value and spread.

1. Identify the model and its parameters. For a normal curve, the center gives μ and the inflection points are at $\mu \pm \sigma$.
2. Substitute into the moment formulas on the summary page. Take the square root of the variance for SD.
3. State the mean as a long-run average and the SD as the scale of variation about it. **Check.** Mean and SD use the variable's units. Variance uses squared units.

Worked example. Compute mean, variance, and SD for the models below.

Time model	Mean	Variance	SD
Uniform labelling $U(3, 15)$	$(3 + 15)/2 = 9$ min	$(15 - 3)^2/12 = 12$ min^2	$\sqrt{12} \approx 3.464$ min
Increasing labelling Tri$(3, 15, 15)$	$(3 + 2(15))/3 = 11$ min	$(15 - 3)^2/18 = 8$ min^2	$\sqrt{8} \approx 2.828$ min
Decreasing production Tri$(1, 10, 1)$	$(2(1) + 10)/3 = 4$ h	$(10 - 1)^2/18 = 4.500$ h^2	$\sqrt{9/2} \approx 2.121$ h
Repair duration Tri$(10, 30, 18)$	$(10 + 30 + 18)/3 \approx 19.333$ min	$(100 + 900 + 324 - 300 - 180 - 540)/18 = 152/9 \approx 16.889$ min^2	$\sqrt{152/9} \approx 4.110$ min
Between repair requests Exp$(12/60)$	$1/(1/5) = 5$ min	$1/(1/5)^2 = 25$ min^2	$1/(1/5) = 5$ min

The increasing labelling model has a higher average time and a smaller SD than the uniform model. Production averages 4 h, repairs average about 19.333 min, and the wait between requests averages 5 min with an SD of 5 min.

Worked example. Infer normal parameters and rescale. Daily revenue R is in thousands of COP. Its center is 1800 and its inflection points are 1680 and 1920, so $\sigma = 1920 - 1800 = 1800 - 1680 = 120$ and $R \sim N(1800, 14400)$. For $Y = kR$, multiply the mean by k, variance by k^2, and SD by $|k|$. With $k = 1000$,

$$E(Y) = 1000(1800) = 1{,}800{,}000 \text{ COP}, \quad \text{Var}(Y) = 1000^2(14400) = 14{,}400{,}000{,}000 \text{ COP}^2,$$
$$\text{SD}(Y) = 1000(120) = 120{,}000 \text{ COP}.$$

The workshop averages 1,800,000 COP per day over the long run, with daily variation measured by the SD of 120,000 COP.

C6 — Estimate an event probability from sample data

Method. Use relative frequency when the question asks how often an event occurred in observed data.

1. Write the event with its endpoints. Count all observations n, and count qualifying observations k. Add counts and totals across disjoint records of the same sample. **Check.** Keep repeated values and apply the endpoint rule to each one.
2. Use $\widehat{P} = k/n$. For a grouped table or frequency histogram, add the counts of the included classes when the cutoffs match class boundaries and the inclusion rules agree.
3. For theoretical probability P_0, calculate $E = |\widehat{P} - P_0|$ and $100E/|P_0|$ as a percentage. Keep the exact fraction during working. **Check.** A zero reference value leaves the percentage error undefined.

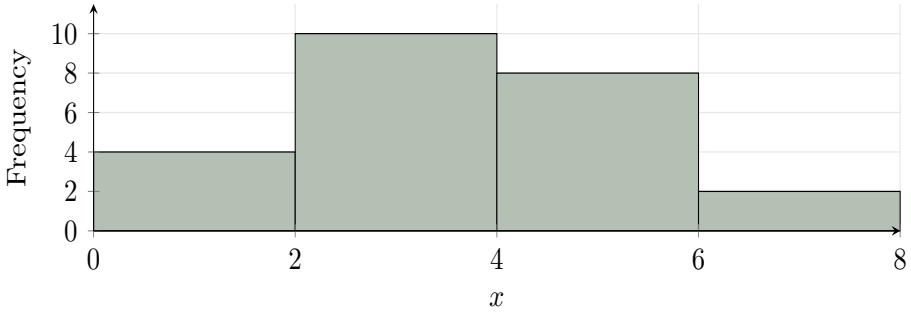
Worked example. Parcel masses in kg, from disjoint records. For $2 \le X < 3$, the eligible values from the raw list of 20 are $2.0, 2.2, 2.4, 2.5, 2.7, 2.0, 2.3, 2.6, 2.9$. In the separate table of 30 parcels, $2.0, 2.4, 2.6, 2.8$ have frequencies $3, 5, 7, 6$. Include 2 and exclude 3.

$$k = 9 + (3 + 5 + 7 + 6) = 30, \qquad n = 20 + 30 = 50, \qquad \widehat{P} = 30/50 = 0.6000.$$

With $P_0 = 0.40$, $E = |30/50 - 0.40| = 0.2000$ and $E/P_0 \times 100\% = 50.00\%$. The sample proportion is 60.00%, or 20 percentage points above the model.

Worked example. Grouped transfer times in seconds. The classes $[0, 4), [4, 8), [8, 12), [12, 16)$ have counts $3, 9, 5, 3$. For $4 \le X < 12$, $k = 9 + 5 = 14$ and $n = 20$, so $\widehat{P} = 14/20 = 0.7000$. Against $P_0 = 0.75$, $E = 0.0500$ and $0.05/0.75 \times 100\% \approx 6.67\%$. The observed share is 70.00%, below the model.

Worked example. Collection-queue wait times from a histogram. Find the observed share with $2 \le X < 6$ min.



Read heights 10 and 8 for the bars from 2 to 6 min. The class on the right starts at 6 min and is excluded.

$$k = 10 + 8 = 18, \qquad n = 4 + 10 + 8 + 2 = 24, \qquad \widehat{P} = 18/24 = 0.7500.$$

With $P_0 = 0.70$, $E = |18/24 - 0.70| = 0.0500$ and $0.05/0.70 \times 100\% \approx 7.14\%$. Three quarters of the sample waited in the stated interval, five percentage points above the model.

C7 — Calculate and compare sample statistics

Method. Use weighted working to find the mean, variance, and SD of a sample.

1. Let v_i be an exact value x_i or, for a grouped record, a midpoint $m_i = (\text{left} + \text{right})/2$. Let f_i be its frequency. A raw entry contributes weight 1. Combine disjoint records by adding frequencies.
2. Calculate $n = \sum f_i$, $\bar{x} = \sum f_i v_i / n$, and $Q = \sum f_i (v_i - \bar{x})^2$. You can also use $Q = \sum f_i v_i^2 - (\sum f_i v_i)^2 / n$.
3. Compute $s^2 = Q/(n-1)$ and $s = \sqrt{s^2}$. **Check.** Use $n-1$ for sample variance. With midpoints, label the statistics and their errors as estimates.
4. For each statistic z requested, compare with its theoretical counterpart z_0 using $E = |z - z_0|$ and $E/|z_0| \times 100\%$ if $z_0 \neq 0$. Keep unrounded working values and match units.

Worked example. Exact cold-room readings in °C. The raw values $6, 8, 9, 11, 12$ combine with frequencies $2, 3, 2$ at $6, 9, 12$. The resulting frequencies at $6, 8, 9, 11, 12$ are $3, 1, 4, 1, 3$.

$$n = 12, \qquad \bar{x} = \frac{3(6) + 8 + 4(9) + 11 + 3(12)}{12} = \frac{109}{12} \approx 9.083,$$
$$Q = \left[3(6^2) + 8^2 + 4(9^2) + 11^2 + 3(12^2)\right] - 109^2/12 = 707/12,$$
$$s^2 = (707/12)/(12-1) = 707/132 \approx 5.356, \qquad s = \sqrt{707/132} \approx 2.314.$$

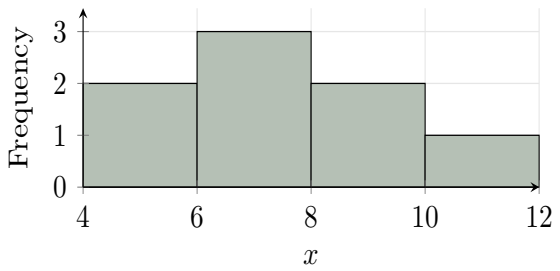
Both $\bar{x}$ and s are in °C, and s^2 is in $(°C)^2$. The mean exceeds the model's 9 by $1/12 \approx 0.083$ °C, or $(1/12)/9 \times 100\% \approx 0.93\%$. Variance exceeds the model's 5 by $47/132 \approx 0.356$ $(°C)^2$, or $(47/132)/5 \times 100\% \approx 7.12\%$.

Worked example. Grouped courier times in minutes. Classes $[8, 12), [12, 16), [16, 20)$ have weights $2, 3, 1$ at midpoints $10, 14, 18$.

$$\bar{x} \approx \frac{2(10) + 3(14) + 18}{6} = \frac{40}{3}, \quad Q = 1112 - 80^2/6 = \frac{136}{3}, \quad s^2 \approx \frac{136/3}{6-1} = \frac{136}{15}.$$

The midpoint estimates are 13.333 min, 9.067 min^2, and $s \approx \sqrt{136/15} \approx 3.011$ min. Against theory 14 min and 12 min^2, both estimates are lower. Their absolute errors are $2/3 \approx 0.667$ min and $44/15 \approx 2.933$ min^2. The percentage errors are $(2/3)/14 \times 100\% \approx 4.76\%$ and $(44/15)/12 \times 100\% \approx 24.44\%$.

Worked example. Container masses from a frequency histogram.



Read midpoints in kg and take the four heights as weights.

Midpoints $5, 7, 9, 11$ have weights $2, 3, 2, 1$, giving $n = 8$. The midpoint working is

$$\bar{x} \approx [2(5) + 3(7) + 2(9) + 11]/8 = 7.500 \text{ kg},$$
$$Q = 480 - 60^2/8 = 30, \qquad s^2 \approx 30/7 \approx 4.286 \text{ kg}^2,$$
$$s \approx \sqrt{30/7} \approx 2.070 \text{ kg}.$$

Against theory 7 kg and 4 kg^2, the estimates are higher by 0.500 kg and $2/7 \approx 0.286$ kg^2. Dividing by 7 and 4 respectively gives 7.14% for both.

C8 — Read the Student's t table

The key uses the signed-deviation model $X = m + aT$, with $a > 0$, $T \sim t_\nu$, and $\nu = n - 1$. Negative X means below the contextual reference. Standardized values are measured relative to m.

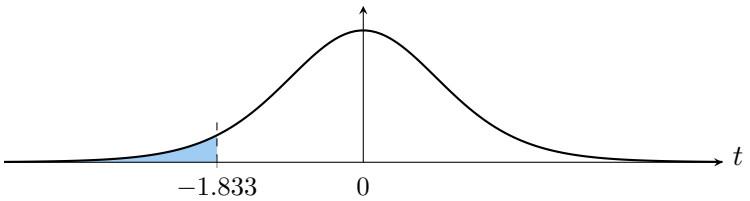
Method. Direct questions give the cutoffs and ask for an area.

1. Write $\nu = n - 1$ and standardize each given cutoff with $c = (x - m)/a$.
2. Find $|c|$ in row ν. The column gives $\alpha = P(T > |c|) = P(T < -|c|)$ at table precision. Use the printed entries $0.30, 0.20, 0.10, 0.05, 0.025, 0.01, 0.005$ for one-tail areas.
3. Reflect a negative cutoff. Complement if the event includes the center and extends to one end. Subtract nested tails for an interval or add disjoint tails. **Check.** At $c = 0$ each tail is 0.5. A tail that includes the center extends over more than half the curve.

The exercise cutoffs match the printed t entries after standardizing. If $|c|$ lies between two entries, the associated columns bound its one-tail probability.

Worked example. Five direct variants. Each (m, a) pair gives the location and scale in the stated units. Use the table areas shown.

Event and givens	Standardize and look up	Combine and interpret
Left tail. Courier arrival deviation $X < -4.499$ min, $(m, a) = (1, 3)$, $n = 10$.	$c = (-4.499 - 1)/3 = -1.833$. Row 9, column 0.05 gives 1.833.	Reflect to the left tail, $P \approx$ 0.05. About 5% arrive more than 4.50 min early.
Right tail. Freight cost deviation $X > -10.984$ dollars, $(m, a) = (-4, 8)$, $n = 13$.	$c = (-10.984 + 4)/8 = -0.873$. Row 12, column 0.20 gives 0.873.	$P \approx 1 - 0.20 =$ 0.80. About 80% cost more than the threshold of 10.98 dollars below budget.
Interval above m. Energy deviation $7.19 < X < 12.476$ kWh, $(m, a) = (2, 6)$, $n = 17$.	$c_L = (7.19 - 2)/6 = 0.865$, $c_U = (12.476 - 2)/6 = 1.746$. Row 16, columns $0.20, 0.05$.	Subtract nested right tails. $P \approx 0.20 - 0.05 = 0.15$. About 15% of days fall in that above-forecast band.
Interval below m. Crop mass deviation $-8.095 < X < -3.688$ tonnes, $(m, a) = (-1, 3)$, $n = 8$.	$c_L = (-8.095 + 1)/3 = -2.365$, $c_U = (-3.688 + 1)/3 = -0.896$. Row 7, columns $0.025, 0.20$.	Subtract the smaller left tail. $P \approx 0.20 - 0.025 = 0.175$. About 17.5% of shipments fall in that below-forecast band.
Unequal outside tails. Revenue alert $X < -5.772$ or $X > 7.76$ in thousands of dollars, $(m, a) = (2, 4)$, $n = 7$.	$c_L = (-5.772 - 2)/4 = -1.943$, $c_U = (7.76 - 2)/4 = 1.440$. Row 6, columns $0.05, 0.10$.	Add disjoint tails. $P \approx 0.05 + 0.10 = 0.15$, so about 15% of weeks trigger an alert.



Courier arrivals. Reflect the printed right-tail cutoff to -1.833. The blue left tail has table area 0.05.

C8 continued — Find cutoffs from areas

Method. Inverse questions give an area and ask for one or two cutoffs.

1. Turn the requested area into the needed one-tail probabilities. For central area p, each excluded tail is $(1-p)/2$. For equal outside tails of total area q, each is $q/2$. Use unequal areas separately.
2. In row $\nu = n-1$, read the positive critical value in each one-tail column. Assign minus to the lower cutoff and plus to the upper cutoff for the small tails shown here. For a tail larger than 0.5, use its complement to identify the cutoff on the opposite side.
3. Back-transform each signed value with $x = m + ac$. Report contextual cutoffs to two decimals. **Check.** For equal tails, the two limits are equally far from m.

Worked example. Five inverse variants using the printed table. The back-transformation puts the limits in the units of X.

Target and givens	Row, column, and sign	Back-transform and interpret
Lower cutoff. Operating cost, dollars, $(m, a) = (-3, 5)$, $n = 15$. Left area 0.10.	Row 14, column 0.10 gives 1.345. Use $c =$ −1.345.	$x = -3 + 5(-1.345) = -9.725 \approx -9.73$. About 10% of days cost more than 9.73 dollars below budget.
Upper cutoff. Loading time, min, $(m, a) = (2, 7)$, $n = 21$. Right area 0.025.	Row 20, column 0.025 gives $c =$ 2.086.	$x = 2 + 7(2.086) = 16.602 \approx$ 16.60. About 2.5% of jobs take more than 16.60 min above target.
Central interval. Demand, tonnes, $(m, a) = (4, 9)$, $n = 12$. Central 0.90.	$(1 - 0.90)/2 = 0.05$. Row 11, column 0.05 gives $c = \pm 1.796$.	$4 \pm 9(1.796)$ gives $(-12.164, 20.164)$. About 90% of demand deviations fall between −12.16 and 20.16 tonnes.
Equal outside tails. Package mass, g, $(m, a) = (-0.4, 1.2)$, $n = 9$. Total 0.05.	$0.05/2 = 0.025$. Row 8, column 0.025 gives $c = \pm 2.306$.	$-0.4 \pm 1.2(2.306)$ gives $(-3.1672, 2.3672)$. About 2.5% fall below −3.17 g and 2.5% above 2.37 g.
Unequal outside tails. Dispatch times, min, $(m, a) = (-2, 6)$, $n = 16$. Left 0.01, right 0.10.	Row 15, column 0.01 gives $c_L = -2.602$ after reflection. Column 0.10 gives $c_U = 1.341$.	$L = -2 - 6(2.602) = -17.612$ and $U = -2 + 6(1.341) = 6.046$. The model puts 1% more than 17.61 min early and 10% more than 6.05 min late.

C9 — Compare normal and Student's t results

For this criterion, choose normal when the population SD is known, and Student's t when the population SD is estimated from a sample, using $\nu = n - 1$. When asked to compare, calculate both.

Shared scale in these comparisons. Use $X_T = m + aT$ and $X_N = m + aZ$ with the same m and stated approximate SD as the scale a. In this teaching convention, $\mathrm{SD}(X_N) = a$ and $\mathrm{SD}(X_T) = a\sqrt{\nu/(\nu-2)}$ for $\nu > 2$. Use the supplied a directly. Table results are approximate because the printed entries are rounded.

Method. For a direct comparison, use the same event and standardize each cutoff once.

1. For t, keep the standardized cutoff c at the printed precision and find the one-tail area in row $n - 1$.
2. For normal, round each c to two decimals to get z before lookup. The table gives $\Phi(z) = P(Z \le z)$. For negative z, use $\Phi(-|z|) = 1 - \Phi(|z|)$. For a right tail use $1 - \Phi(z)$, and for an interval use $\Phi(z_U) - \Phi(z_L)$.
3. For results r_T, r_N, report $E_A = |r_T - r_N|$, then $E_{Pt} = E_A/|r_T| \times 100\%$ and $E_{Pn} = E_A/|r_N| \times 100\%$. **Check.** Use four decimals for probabilities and their absolute difference, two for the percentages. A zero denominator leaves the corresponding percentage undefined.

Worked example. The table pairs the methods for three events. Each comparison uses the same (m, a) and sample size.

Event and cutoffs	Student's t	Normal
Sales shortfall $X < -11.165$ thousand dollars. $(m, a) = (-2, 5)$, $n = 10$. $c = (-11.165 + 2)/5 = -1.833$.	Row 9, column 0.05, gives 1.833. Reflect for $p_T =$ 0.0500.	$z =$ −1.83. Row 1.8, column 0.03, gives $\Phi(1.83) =$ 0.9664. $p_N = 1 - 0.9664 =$ 0.0336.
Delivery deviation $X > -0.472$ min. $(m, a) = (3, 4)$, $n = 15$. $c = (-0.472 - 3)/4 = -0.868$.	Row 14, column 0.20, gives 0.868. The requested area is $p_T = 1 - 0.2000 = 0.8000$.	$z = -0.87$. Row 0.8, column 0.07, gives $\Phi(0.87) = 0.8078$. $p_N = 1 - (1 - 0.8078) = 0.8078$.
Fuel deviation $-7.136 < X < 14.074$ L. $(m, a) = (1, 6)$, $n = 13$. $c_L = (-7.136 - 1)/6 = -1.356$, $c_U = (14.074 - 1)/6 = 2.179$.	Row 12, columns 0.10 and 0.025, give the two outside areas. $p_T = 1 - 0.1000 - 0.0250 = 0.8750$.	$z_L = -1.36$, $z_U = 2.18$. Row 1.3, col. 0.06 gives 0.9131. Row 2.1, col. 0.08 gives 0.9854. $p_N = 0.9854 - (1 - 0.9131) = 0.8985$.

Event	E_A	E_{Pt}	E_{Pn}
Sales shortfall	$\lvert 0.0500 - 0.0336\rvert = 0.0164$	$0.0164/0.0500 \times 100\% = 32.80\%$	$0.0164/0.0336 \times 100\% \approx 48.81\%$
Delivery	$\lvert 0.8000 - 0.8078\rvert = 0.0078$	$0.0078/0.8000 \times 100\% \approx 0.98\%$	$0.0078/0.8078 \times 100\% \approx 0.97\%$
Fuel band	$\lvert 0.8750 - 0.8985\rvert = 0.0235$	$0.0235/0.8750 \times 100\% \approx 2.69\%$	$0.0235/0.8985 \times 100\% \approx 2.62\%$

Student's t gives the sales shortfall more probability by 1.64 percentage points. Normal gives the delivery event 0.78 percentage points more and the fuel band 2.35 percentage points more. The percentages compare model results with each other.

C9 continued — Compare inverse cutoffs

Method. For an inverse comparison, start with areas and find a cutoff under each model.

1. Write the one-tail areas for t and the left cumulative targets for normal. A right-tail area α gives normal target $1-\alpha$.
2. Read the signed t cutoff from the one-tail columns. Find the closest printed normal CDF entry, using symmetry for negative cutoffs. For a tie at 0.0500 or 0.9500, choose $|z| = 1.64$.
3. Back-transform each standardized cutoff using $x = m + ac$. Compare unrounded contextual results, pairing lower with lower and upper with upper. **Check.** Round cutoffs and their absolute differences to two decimals with units. Give percentage differences to two decimals.

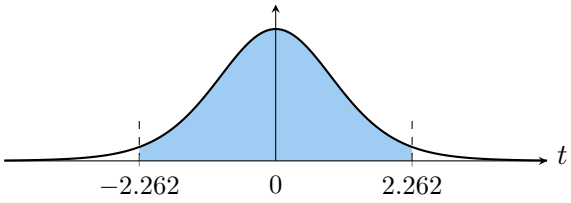
Worked example. Single-cutoff variants.

Event and givens	**Student's t**	**Normal**
Lowest 5% of batch cost deviations, dollars. $(m, a) = (-1, 2.5)$, $n = 8$.	Row 7, column 0.05, gives $c_T = -1.895$ by symmetry. $x_T = -1 + 2.5(-1.895) = -5.7375$.	Row 1.6, columns $0.04, 0.05$, gives $0.9495, 0.9505$. The reflected areas $0.0505, 0.0495$ tie, so $c_N = -1.64$ and $x_N = -1 + 2.5(-1.64) = -5.10$.
Highest 10% of order processing deviations, min. $(m, a) = (2, 3.5)$, $n = 17$.	Row 16, column 0.10, gives $c_T = 1.337$. $x_T = 2 + 3.5(1.337) = 6.6795$.	Target 0.9000. Row 1.2, column 0.08, gives 0.8997, closer than 0.9015 in column 0.09. Use $c_N =$ 1.28, so $x_N = 2 + 3.5(1.28) =$ 6.48.

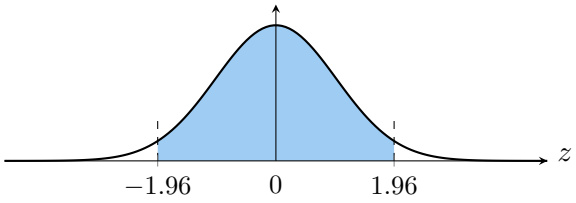
The batch savings thresholds are 5.74 dollars under t and 5.10 dollars under normal. From the unrounded cutoffs, $E_A = 0.6375 \approx 0.64$ dollars, $E_{Pt} = 0.6375/5.7375 \times 100\% \approx 11.11\%$, and $E_{Pn} = 0.6375/5.10 \times 100\% = 12.50\%$.

For the slowest orders, the upper cutoffs are 6.68 and 6.48 min above target. Here $E_A = 0.1995 \approx 0.20$ min, $E_{Pt} = 0.1995/6.6795 \times 100\% \approx 2.99\%$, and $E_{Pn} = 0.1995/6.48 \times 100\% \approx 3.08\%$.

Worked example. Central 95% of cargo mass deviations in kg. Use $(m, a) = (1.5, 4)$, $n = 10$, and split the remaining 0.05 into two tails of 0.025. Row 9, column 0.025, gives $c_T = \pm$ 2.262. Normal row 1.9, column 0.06, gives $\Phi(1.96) =$ 0.9750, so the cutoffs for targets $0.0250, 0.9750$ are $c_N = \pm$ 1.96.



Student's t, df = 9. Read the central 95% between the two t cutoffs.



Standard normal. Read the same central area between the closer normal cutoffs.

End	**Back-transform t**	**Back-transform normal**	E_{Pt}	E_{Pn}
Lower	$1.5 - 4(2.262) = -7.548$	$1.5 - 4(1.96) = -6.34$	$100(1.208)/7.548 \approx 16.00\%$	$100(1.208)/6.34 \approx 19.05\%$
Upper	$1.5 + 4(2.262) = 10.548$	$1.5 + 4(1.96) = 9.34$	$100(1.208)/10.548 \approx 11.45\%$	$100(1.208)/9.34 \approx 12.93\%$

Each endpoint differs by $1.208 \approx 1.21$ kg in absolute value. The central cargo deviations are about $(-7.55, 10.55)$ kg under t and $(-6.34, 9.34)$ kg under normal. In both cases, 2.5% lie beyond each end.

C10 — Choose a model and plan an investigation

Method. For the brief exam response, choose one of the proposed models and describe plausible data in one or two sentences.

1. Match the variable to one option. Consider a linear density over a finite interval, an exponential distribution for waiting times, or a normal distribution for measurements.
2. Name a real economic or financial setting, the variable, and its units. Explain the condition that makes your proposed model plausible.
3. State what you would check with observations. **Check.** Describe model fit as something to investigate with data.

Worked example. Exponential waiting times. *I would study the minutes between payment requests reaching a small business during a stable trading period, proposing an exponential model if requests arrive independently at a constant average rate. I would use real observations to check those assumptions.*

Separate report checklist

Use your selected model to investigate an economic or financial question.

- ☐ **Research question.** State the question, the variable you will study, and its economic or financial context.
- ☐ **Real data.** Identify the observations you used and explain what they measure.
- ☐ **Model and methods.** Justify the selected continuous model and the analysis methods.
- ☐ **Simulation.** Describe the simulation and report its results.
- ☐ **Sampling analysis.** Present the sampling analysis and explain what the results show.
- ☐ **Portfolio risk.** Include and interpret the portfolio-risk measures used.
- ☐ **Assumptions and limitations.** State the conditions your analysis relies on and explain its limits.
- ☐ **Conclusion.** Answer the research question with evidence from your results.

Return to the C1–C10 summary